Project Brief Form 2025-26

Student Name	**KUZEY ERTURK**	*Id Number*	**F330035**
Supervisor	**PROF. NINA DETHLEFS**		
Programme	**Computer Science BSc**	*Module code*	**25COC251**

ALL BOXES IN THIS SECTION SHOULD BE TICKED.

☒ The attached project brief is agreed between the supervisor and the student as a description of the proposed project work.

Bottom of Form

☒ A provisional workplan is attached (on subsequent pages.)

☒ The Ethical Awareness Form (EAF) has been filled and attached (on subsequent pages.)

☒ If any answers on the EAF have been marked as **Yes** (except for the first), then a formal application has to go through LEON (leon.lboro.ac.uk).

Part C students to highlight/mark the column matching their project module code (remove other X's).
Part D (COD290) students to highlight the column matching their part C project module code.

Project Code --> Project Title --> Programme(s) -->	COC251 CS (CS)	COC252 Computing (C+M, ITMB)	COC253 IT (C+M, ITMB)	COC255 CS+Math (CS-Math)	COC257 CS+AI (CS-AI)
Requires CS content.	X				
Requires maths content.					
Requires AI content.					
Requires IT content.					
Requires a real customer.					
Requires Novelty					

Bottom of Form

Students' Initials:	KE

Top of Form

Loughborough University

Ethics Awareness Form for Taught Student Projects

Project Title/Topic: AI-Driven Crop Yield Prediction System

All students should discuss with their supervisor whether their project might conflict with the University's ethical principles which can be found in the Ethical Policy Framework.

Students should complete the second column in the table below, discussing with their supervisor as appropriate.

Aspect of project	**Does the project involve this aspect? (Yes / No)**	**If Yes, follow the process(es) below**
Analysis of secondary or pre-existing data which **does not** require ethical review based on the Guidance Note: Studies Using Secondary or Pre-Existing Data	Yes	No further ethical review required.
Analysis of secondary or pre-existing data which **does** require ethical review based on the Guidance Note: Studies Using Secondary or Pre-Existing Data	No	Complete an ethics application via LEON https://leon.lboro.ac.uk For submission guidance see: https://www.lboro.ac.uk/internal/research-ethics-integrity/research-ethics/ethical-review/leon/
Investigations involving human participants	No	Follow the Code of Practice on Investigations involving Human Participants. Complete an ethics application via LEON https://leon.lboro.ac.uk For submission guidance see: https://www.lboro.ac.uk/internal/research-ethics-integrity/research-ethics/ethical-review/leon/
Investigations involving activity falling under the Human Tissues Act	No	Follow guidance from the Human Tissue Act Licence Sub-Committee If necessary, complete an ethics application via LEON https://leon.lboro.ac.uk For submission guidance see: https://www.lboro.ac.uk/internal/research-ethics-integrity/research-ethics/ethical-review/leon/

V2.1 (V1 approved by Learning and Teaching Committee)

Investigations with military applications or using dual use technologies	No	Complete the Review Process for Projects Involving Research with Military Applications or Dual Use Technologies.
Investigations involving animals or animal cells/tissues	No	Complete an ethics application via LEON https://leon.lboro.ac.uk For submission guidance see: https://www.lboro.ac.uk/internal/research-ethics-integrity/research-ethics/ethical-review/leon/
Investigations involving accessing security sensitive material (e,g, online terrorist content or material).	No	Complete an ethics application via LEON https://leon.lboro.ac.uk For submission guidance see: https://www.lboro.ac.uk/internal/research-ethics-integrity/research-ethics/ethical-review/leon/
Possible conflict with ethical principles partially or wholly outside the above.	No	Forward a study description to the Dean of School or the Responsible Person (see Ethical Policy Framework for list)

Student Declaration

I confirm that I have discussed the ethics awareness form with my supervisor and, if appropriate, followed the relevant guidance / made the relevant application.

Student name: Kuzey Erturk

Student ID number: F330035

Date: 22/10/2025

Supervisor Declaration

I confirm that I have discussed the ethics awareness form with my supervisee and, if appropriate, requested that they follow the relevant guidance / make the relevant application.

Supervisor name: Nina Dethlefs

Date: 23/10/2025

Loughborough University

Bottom of Form

Project Description

Project Title

AI-Driven Crop Yield Prediction System

Background and Motivation

Productivity in agriculture nowadays faces increasing challenges from climate variability to biological factors. The rapid ongoing increase in population and increasing food demand necessitate efficient agricultural practices to enable people to manage limited natural resources sustainably. Climate change causes extreme weather events such as droughts and floods, creating greater uncertainty in crop production cycles. Accurately predicting the crop yield can enable farmers to optimize their resource allocation and mitigate the risks. Traditional methods rely on simple statistical approaches that fail to capture complex factors and interactions between environmental factors and crop performance. Machine learning offers the potential to model these relationships and provide more accurate predictions by learning from historical patterns. By leveraging historical yield records with meteorological and management data, these models can provide insights that can increase agricultural resilience and productivity.

Project Aim

To develop and evaluate the machine learning models that predict the crop yield using historical data, agricultural management factors and patterns.

Project Objectives

1. Compile and integrate multi-source agricultural datasets covering crop yields and environmental factors
2. Perform analysis to identify relationships between meteorological conditions and yield outcomes
3. Evaluate the features such as temporal trends, environmental ordinaries and characteristics of crops that will affect the outcome
4. Develop and compare different machine learning models including baseline and ensemble methods
5. Evaluate the performance of the model and assess each different predictor and their relative importance
6. Compare prediction across different crop types and evaluate the patterns between the crops

Methodology

The project will follow a systematic machine learning workflow:

Dataset Collection and Integration**:** Multiple datasets will be combined that will consist of factors that have affect on crop yield such as rainfall ,temperature etc. Data sources will be validated for accuracy and temporal alignment to ensure consistency.

Data Preprocessing and Feature Engineering: Raw data transformed into features suitable for machine learning by creating additional variables that can capture patterns, combinations of environmental conditions that will affect our prediction of crop yields

Model Development: Multiple machine learning algorithms will be implemented, starting with baseline models to be able to compare the advanced models with them. Advanced models including Random Forests, Gradient Boosting, and potentially Neural Networks will then be developed and trained on historical data.

Model Evaluation and Validation: Models will be evaluated using regression metrics such as RMSE, MAE, and R-squared values.

Comparative Analysis: Model performance will be compared across different algorithms and crop types. Feature importance analysis will identify which factors most strongly influence yield predictions, providing interpretable insights for decision-making.

Expected Deliverables

- A progress report on the project - Wednesday, Week 12 after Christmas break
- Hand in the project - 21/04/2025
- Project demo / presentation - wc 27/04/2025

Success Criteria

This project will be successful and reach its goal if it provides a model that outperforms baseline predictions, identifies statistically significant environmental relationships, provides interpretable insights, demonstrates differences across crop types and delivers well documented analysis.

Workplan with Gantt Chart